**Instructions:**

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

**Q. 1:** Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=5;i++)
    (i-1)[arr] = (i%5);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 3;
}
int main(){
  int arr[6] = {2,2,2,2,2};
  func1(arr+1);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

1 2 3 4 0

arr 0 1 2 3 4

2 1 2 3 4 0

0 1 2 3

arr[5] = 0

Address of a[0]

arr = a[0]

| Output |
|---|
| 2 1 2 3 4 0 |
| 3 1 2 3 40 |
| Will the answer change if we write func2 as:<br>`void func2 (int a[1] , int arr){`<br>`int temp = arr + 4;`<br>`a = &temp;`<br>`}`<br>Explain your answer |
| ~~Yes the~~<br>No the Answer will not change. |

**Q. 2:** Write the output of the following sequence of instructions **long arr2[90]; int arr1[80], a1=(&arr2[60]), a2=(&arr2[10]), b1=(&arr1[60]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

4 Addres

Address

8

400 200

For long the box size is 8 bytes and for int it is 4 bytes.
$a_1, a_2, b_1$ & $b_2$ are address locations of arr2 & arr1 elements.
Also they are stored contiguously Hence the total difference is printed.